

Ethics ETH2526-1929: Mr Idowu Afolabi Isaac Ojo (Low risk)

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Student ID	
Category	Doctoral Researcher Doctoral Researcher Academic Staff
Project	Graphical Perception of Position and Angle: A Replication and Follow-up of Cleveland & McGill (1984)
School	School of Science & Technology
Department	Department of Computer Science
Current status	Approved

Ethics application

Risks

R1) Does the project have funding?

Yes

R2) Does the project involve human participants?

Yes

R3) Will the researcher be located outside of the UK during the conduct of the research?

No

R4) Will any part of the project be carried out under the auspices of an external organisation, involve collaboration between institutions, or involve data collection at an external organisation?

Yes

R5) Does your project involve access to, or use of, terrorist or extremist material that could be classified as security sensitive?

No

R6) Does the project involve the use of live animals?

No

R7) Does the project involve the use of animal tissue?

No

R8) Does the project involve accessing obscene materials?

No

R9) Does the project involve access to confidential business data (e.g. commercially sensitive data, trade secrets, minutes of internal meetings)?

No

R10) Does the project involve access to personal data (e.g. personnel or student records) not in the public domain?

No

R11) Does the project involve deviation from standard or routine clinical practice, outside of current guidelines?

No

R12) Will the project involve the potential for adverse impact on employment, social or financial standing?

No

R13) Will the project involve the potential for psychological distress, anxiety, humiliation or pain greater than that of normal life for the participant?

No

R14) Will the project be conducted or supported by any U.S. federal department or agency?

No

R15) Will the project involve research into illegal or criminal activity where there is a risk that the researcher will be placed in physical danger or in legal jeopardy?

No

R16) Will the project specifically recruit individuals who may be involved in illegal or criminal activity?

No

R17) Will the project involve engaging individuals who may be involved in terrorism, radicalisation, extremism or violent activity and other activity that falls within the Counter-Terrorism and Security Act (2015)?

No

Applicant & research team

T1) Principal Applicant

Name

[Mr Idowu Afolabi Isaac Ojo](#)

T2) Co-Applicant(s) at City

Name

[Mrs Sabah Azhar](#)

Name

[Prof Jason Dykes](#)

T3) External Co-Applicant(s)

Name

Cagatay Turkey

Title of post

Professor

Affiliation

University of Warwick

Provide details of the researcher's insurance cover.

Covered by Warwick's Professional Indemnity Policy, which covers "the Legal Liability of the University to Third Parties for the breach of Professional duty due to the negligence of the University in the course of its business."

Name

Fiona Chung

Title of post

Research Student

Affiliation

University of Warwick

Provide details of the researcher's insurance cover.

Covered by Warwick's Professional Indemnity Policy, which covers "the Legal Liability of the University to Third Parties for the breach of Professional duty due to the negligence of the University in the course of its business."

T5) Do any of the investigators have direct personal involvement in the organisations sponsoring or funding the research that may give rise to a possible conflict of interest?

No

T6) Will any of the investigators receive any personal benefits or incentives, including payment above normal salary, from undertaking the research or from the results of the research above those normally associated with scholarly activity?

No

T7) List anyone else involved in the project.

Project details

P1) Project title

Graphical Perception of Position and Angle: A Replication and Follow-up of Cleveland & McGill (1984)

P1.1) Short project title

Graphical Perception of Position and Angle

P2) Provide a lay summary of the background and aims of the research, including the research questions (max 400 words).

When designing graphs, it's important to consider how accurately viewers will be able to extract numeric information from the chosen graphical elements. The more accurately viewers can decode graphs, the better they will be able to understand the data and the patterns being presented. To establish a scientific basis for graphical perception, Cleveland & McGill (1984) published a foundational theory outlining a set of elementary perceptual tasks that viewers carry out when extracting quantitative information from graphs, and ordered these tasks based on how accurately viewers perform them.

One of these tasks is judging position along a common scale, such as comparing the height of bars in a bar chart. Another is judging angle, such as comparing segments of a pie chart. Cleveland & McGill (1984) suggested that judgements of position along a common scale were more accurate than judgements of angle when extracting quantitative information from graphs. They supported this theory through empirical testing. However, they only tested this claim in the context of making part-to-part comparisons (i.e., comparing individual bars or segments to each other).

Making part-to-whole comparisons (i.e., comparing individual bars or segments to the overall total of all bars or segments) is another common task viewers may have to conduct when examining graphs (Kosara, 2019). This task is much more commonly used for pie charts than making part-to-part comparisons, which may be why participants performed poorly when making part-to-part judgements in the original Cleveland & McGill (1984) experiment (Hill, 2024).

Judgements of angle may not be significantly less accurate than judgements of position when making part-to-whole comparisons. This is important because the original paper claims that bar charts should always be used in favor of pie charts. However, if viewers perform part-to-whole comparisons more accurately with pie charts, there may be cases where pie charts are more appropriate for communicating information than bar charts.

We aim to (1) replicate the Position-Angle Experiment reported in Cleveland & McGill (1984), which compared estimation accuracy for bar charts and pie charts during part-to-part comparison tasks, and (2) extend this experiment in a follow-up study where we introduce a part-to-whole comparison task that was not included in the original study.

We intend to address the following research question: Can we replicate Cleveland and McGill's original claim that position judgements lead to more accurate estimation of quantitative information than angle judgements in both part-to-part and part-to-whole comparison tasks?

References:

Cleveland, W. S., & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.2307/2288400>.

Kosara, R. (2019). Evidence for Area as the Primary Visual Cue in Pie Charts. *IEEE Visualization Conference (VIS)*, 101-105. <https://doi.org/10.1109/VISUAL.2019.8933547>.

Hill, A. (2025). Are pie charts evil? An assessment of the value of pie and donut charts compared to bar charts. *Information Visualization*, 24(1), 3-23.

P4) Provide a summary and brief explanation of the research design, method, and data analysis.

The study will employ a 2×2 mixed factorial design, with encoding type (position vs. angle) manipulated within participants and comparison type (part-to-part vs. part-to-whole) manipulated between participants.

Independent Variable 1: encoding type, position versus angle, within-subjects.

Independent Variable 2: comparison type, part-to-part versus part-to-whole, between-subjects.

Dependent Variable: judged percentage.

Participants will be allocated to one of two comparison type conditions: the part-to-part condition or the part-to-whole condition. All participants will see both encoding types - bar charts and pie charts. Therefore, encoding type (position vs angle) will be manipulated within participants, while comparison type (part-to-part versus part-to-whole) will be manipulated between participants.

In the part-to-part condition, participants will be presented with a sequence of 10 bar charts and 10 pie charts in a randomised order. These charts will be created specifically for this study by the researchers using simulated data with exactly 5 bars or 5 segments, in line with Cleveland and McGill's methodology. Participants will then be asked to estimate what percentage each of the other bars or segments is of the largest one. They will be asked to make quick visual judgements, rather than precise measurements. Each participant will make four judgements for each chart, resulting in 40 position judgements and 40 angle judgements, totaling 80 judgements.

In the part-to-whole condition, participants will be presented with the same type of sequence: 10 bar charts and 10 pie charts in a randomised order. However, they will be asked to estimate what percentage an individual bar or segment is of the whole chart. Again, they will be asked to make quick visual judgements, rather than precise measurements. Each participant will make four judgements for each chart, resulting in 40 position judgements and 40 angle judgements, totaling 80 judgements.

For both conditions, participants will complete a practice trial to ensure they understand the task. They will have the option to take a short break halfway through the study.

See protocol diagram in P4.1

P4.1) If relevant, please upload your research protocol.

P5) What do you consider are the ethical issues associated with conducting this research and how do you propose to address them?

Informed consent: Participants will receive an online information page before taking part and will provide digital consent before beginning the study. Participants can stop participating at any point during the survey by exiting the webpage.

Though this is a low risk study, the main possible burdens are mild eye strain, fatigue, or frustration from making visual estimates. To reduce these risks, the study is kept short, designed to take approximately 20 minutes on average, and no longer than 30 minutes. Participants will also have the option to take a short break halfway through the study. The participant information page also clearly states that we are interested in how people interpret charts visually, and not in assessing or scoring individual participants.

Data protection: We will collect the minimum data needed to run and analyse the study. All data will be stored securely in line with organisational and legislative requirements. Only the researchers and supervisors will have access to the data. Personal information, including Prolific IDs, will not be shared beyond City St George's and will only be accessed by those studying at or employed by City St George's who are listed as applicants on this study. Additionally, payment will be facilitated through Prolific. The researchers will not have access to any financial information. The dataset will be fully anonymised by removing Prolific IDs once payments have been made. The fully anonymised dataset, with Prolific IDs removed, will be made publicly available through a research repository. Collaborators from the University of Warwick may access fully anonymised research data only.

Fairness: Participants will be paid for their engagement in line with the National Living Wage.

Accessibility barriers: Due to the nature of the study, we will specify that participants who do not have normal or corrected-to-normal vision will not be eligible to take part.

P6) Project start date

The start date will be the date of approval.

P7) Anticipated project end date

01 Oct 2026

P8) Where will the research take place?

The research will be conducted remotely in the UK with UK-based participants. Participants will sign up via Prolific, where they will be directed to the City Qualtrics server.

We will use City St George's information systems with data stored on secure servers within the EU in line with GDPR.

P10) Is this application or any part of this research project being submitted to another ethics committee, or has it previously been submitted to an ethics committee?

No

Funding

F1) Funder

Engineering and Physical Sciences Research Council (EPSRC), Grant Reference: EP/Y035909/1

F2) Does the funder require external membership on the approving REC?

No

F3) Has the funding been approved?

Yes

F4) Value of grant

£ 800

External organisations

E1) Provide details of the external organisation/institution involved with this project.

Collaborators from the University of Warwick are involved in accessing the fully anonymised research data. Personal information will not be shared between institutions.

E2) If applicable, has permission to conduct research in, at or through another institution or organisation been obtained?

No

Human participants: information and participation

The options for the following question are one or more of:

'Under 18'; 'Adults at risk'; 'Individuals aged 16 and over potentially without the capacity to consent'; 'None of the above'.

H1) Will persons from any of the following groups be participating in the project?

None of the above

H2) How many participants will be recruited?

134

H3) Explain how the sample size has been determined.

The original experiment had an initial sample of 55 then after excluding 4, were left with a final sample size of $N = 51$, so this sample size should have sufficient power to detect the same effect.

We aim to recruit approximately 20% more than the original sample to allow exclusions due to incomplete responses, or not following instructions for example. This means we will have a total of 124 participants, 62 per experiment.

For a pilot study, we aim to recruit an additional 10 participants (Bujang et al., 2024).

Note: the data collected from these participants may be added to the final study if no changes are made to the study design. However, if any changes are made, this data will be excluded to avoid polluting the response pool.

References:

Bujang, M. A., Omar, E. D., Foo, D. H. P., & Hon, Y. K. (2024). Sample size determination for conducting a pilot study to assess reliability of a questionnaire. *Restorative dentistry & endodontics*, 49(1), e3. <https://doi.org/10.5395/rde.2024.49.e3>

H4) What is the age group of the participants?

Lower Upper

18

H5) Please specify inclusion and exclusion criteria.

Participants invited to the study must meet the study eligibility criteria:

- be aged 18 or over;
- be fluent in English;
- have normal or corrected-to-normal vision
- use a desktop or laptop computer rather than a mobile phone or tablet;
- be registered on Prolific;
- be a resident in the UK;
- be able to complete a short online visual task.

Participants will be excluded if:

- they use a mobile phone to complete the experiment (smaller screens may affect visual judgement accuracy and stimulus presentation consistency).
- they don't provide consent
- they withdraw before completion
- have incomplete responses
- do not follow instructions
- they have participated in similar studies (i.e. other Cleveland & McGill replication studies)

Note: Participants who begin the study but withdraw before completion, do not wish to consent or do not wish for their data to be collected or analysed, will still receive payment. If participants select these options when asked for consent at the end of the study, they will be redirected to Prolific where their entry will be logged for payment and they will receive a completion code.

H6) What are the potential risks and burdens for research participants and how will you minimise them?

Though this is a low risk study, the main possible burdens are mild eye strain, fatigue, or frustration from making visual estimates. To reduce these risks, the study is kept short, designed to take approximately 20 minutes on average, and no longer than 30 minutes. Participants will also have the option to take a short break halfway through the study. The participant information page also clearly states that we are interested in how people interpret charts visually, and not in assessing or scoring individual participants.

H7) Will you specifically recruit pregnant women, women in labour, or women who have had a recent stillbirth or miscarriage (within the last 12 months)?

No

H8) Will you directly recruit any staff and/or students at City?

None of the above

H8.1) If you intend to contact staff/students directly for recruitment purpose, please upload a letter of approval from the respective School(s)/Department(s).

H9) How are participants to be identified, approached and recruited, and by whom?

Participants can be recruited via the online panel provider: Prolific. Any special considerations in line with the inclusion and exclusion criteria detailed above (H5), will be specified on Prolific.

Recruitment will take place on a voluntary basis. Individuals who participate will be presented with a Participant Information Page and Consent Page before beginning the study.

Participants may withdraw from the study at any point prior to submission without penalty.

H10) Please upload your participant information sheets and consent form, or if they are online (e.g. on Qualtrics) paste the link below.

This 'Participant Information Page' is based on the University templates, our discussion following the initial submission, and adapted for the online context.

H11) If appropriate, please upload a copy of the advertisement, including recruitment emails, flyers or letter.

H12) Describe the procedure that will be used when seeking and obtaining consent, including when consent will be obtained.

Participants will be recruited through Prolific and directed to the online study platform: Qualtrics.

Before starting, they will receive a Participant Information Page explaining:

- the purpose of the research
- why they have been invited to take part
- if they have to take part
- what will happen if they take part
- what the benefits and risks are of taking part
- what data will be collected
- data privacy
- who has reviewed the research project
- data privacy details
- contact details if there was a problem
- further information and contact details for the research team

Consent will be obtained electronically before the study through an online consent form, and at the end of the study.

Once participants have read the Participant Information Page, and agree to take part, they will access the study and be asked to complete it online. At the end of the study, participants will

complete a final consent form where they will confirm if they are happy for their data to be collected and analysed for this study. Here, if participants do not wish to consent, they will be able to close the browser or select the option where they will be redirected to Prolific.

H13) Are there any pressures that may make it difficult for participants to refuse to take part in the project?

No

H14) Is any part of the research being conducted with participants outside the UK?

No

Human participants: method

The options for the following question are one or more of:

'Invasive procedures (for example medical or surgical)'; 'Intrusive procedures (for example psychological or social)'; 'Potentially harmful procedures of any kind'; 'Drugs, placebos, or other substances administered to participants'; 'None of the above'.

M1) Will any of the following methods be involved in the project:

None of the above

M2) Does the project involve any deceptive research practices?

No

M3) Is there a possibility for over-research of participants?

No

M4) Please upload copies of any questionnaires, topic guides for interviews or focus groups, or equivalent research materials.

M5) Will participants be provided with the findings or outcomes of the project?

No

M6) If the research is intended to benefit the participants, third parties or the local community, please give details.

M7) Are you offering any incentives for participating?

Yes

M7.1) Please give details, justifying their type and amount.

Participants will be reimbursed £4.24. This is based on the National Living Wage of £12.71 per hour at time of writing, which amounts to approximately £4.24 for 20 minutes of participation.

M8) Does the research involve clinical trial or clinical intervention testing that does not require Health Research Authority or MHRA approval?

No

M9) Will the project involve the collection of human tissue or other biological samples that does not fall under the Human Tissue Act (2004) that does not require Health Research Authority Research Ethics Service approval?

No

M10) Will the project involve potentially sensitive topics, such as participants' sexual behaviour, their legal or political behaviour, their experience of violence?

No

M11) Will the project involve activities that may lead to 'labelling' either by the researcher (e.g. categorisation) or by the participant (e.g. 'I'm stupid', 'I'm not normal')?

No

Data

D1) Indicate which of the following you will be using to collect your data.

Questionnaire

Computer-based tasks, screen recording or software instrumentation

D2) How will the the privacy of the participants be protected?

De-identified samples or data

D3) Will the research involve use of direct quotes?

No

D5) Where/how do you intend to store your data?

Storage at City

Storage at other site

D5.1) If stored at another site, please provide details.

Storage on City information systems (Qualtrics, One Drive) prior to open access sharing of anonymised data as described below.

D6) Will personal data collected be shared with other organisations?

No

D7) Will the data be accessed by people other than the named researcher, supervisors or examiners?

Yes

D7.1) Explain by whom and for what purposes.

D6 Qualification:

City St George's researchers will not share personal data with University of Warwick collaborators, but draw upon the "least privilege" principle of GDPR and a successful protocol used by Warwick in other collaborations, as follows.

The questionnaire will be hosted on Qualtrics at City.

It will be accessed by Warwick collaborators through their Warwick accounts / emails.

The study itself will be (has been) set up collectively.

Once the study is released, prior to payments being made, during which personal data (Prolific IDs) are collected, Warwick collaborators will be denied access to the collected data through Qualtrics (view permissions removed as documented by Qualtrics, and confirmed by our own investigations and testing).

Payments to participants will be made.

When all responses have been acquired and payments have been made, the data will be anonymised, with Prolific IDs securely removed.

The anonymous data will be shared with Warwick collaborators for analysis.

Qualtrics Anonymisation Workflow is:

1. Data will be exported from Qualtrics to secure City St George's systems.
2. Identifiers, including Prolific IDs, will be removed from the exported dataset.
3. The original identifiable/pseudonymised export will be securely deleted once payment processing has been completed.
4. The cleaned dataset will be checked to ensure it no longer contains Prolific IDs or other direct participant identifiers.

Only the fully anonymised dataset will be used for analysis and shared with University of Warwick collaborators.

5. File deletions will be undertaken in accordance with City St George's University of London's data protection policies.

The anonymised data will be made available for scrutiny through an Open Access server (e.g Figshare) in line with EPSRC and institutional guidance.

We are in the process of completing a Data Management Plan and following guidance from the Research Office will make the data available under a non-commercial attribution license: CC BY-NC 4.0 <https://creativecommons.org/licenses/by-nc/4.0/>

D8) Is the data intended or required (e.g. by funding body) to be published for reuse or to be shared as part of longitudinal research or a different/wider research project now or in the future?

Yes

D9) Does the funding body or your professional organisation/affiliation place obligations or recommendations on the retention and destruction of research data?

Yes

D9.1) What are your affiliations/funding and what are the requirements?

UKRI/EPSC and both institutions (City St George's and The University of Warwick) advocate for Open data provision where possible.

Our intention is to provide anonymised data for scrutiny on a secure server and our approach will enable this (see D7.1).

D10) How long are you intending to keep the research data generated by the study?

Minimum of 10 years

D11) How long will personal data be stored or accessed after the study has ended?

Prolific IDs will be captured using URL parameters in Qualtrics and kept until payment is processed, after which they will be destroyed in accordance the above with City St George's University of London's data protection policies and the Qualtrics Anonymisation Workflow described above (D7.1).

Personal data will not be stored or accessed after the study has ended. This will be securely deleted following payment as described in our Anonymization Workflow (D7.1)

D12) How are you intending to destroy the personal data after this period?

Via ServiceNow, which has a Data Destruction service.

Health & safety

HS1) Are there any health and safety risks to the researchers over and above that of their normal working life?

No

HS3) Are there hazards associated with undertaking this project where a formal risk assessment would be required?

No